

WHEN MORE DATA HURTS: DISTRIBUTIONAL OVERSMOOTHING IN CONTINUAL LEARNING

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ABSTRACT

Deep learning models deployed in real-world applications often encounter new data distributions over time. While continual learning research focuses on mitigating catastrophic forgetting, we identify a more subtle failure mode: *distributional oversmoothing*. We hypothesize that when models sequentially learn from data spanning distinct distributions, decision boundaries become overly smoothed, degrading performance on previously learned tasks even when forgetting is minimized. Through controlled experiments on benchmark datasets, we demonstrate how adding diverse data paradoxically reduces accuracy on original distributions. Analyzing representation geometry reveals increased inter-class confusion and loss of discriminative features, challenging the assumption that more data always improves generalization. We propose diagnostic metrics to identify oversmoothing and preliminary strategies for mitigation, highlighting its implications for deploying models in evolving environments.

1 INTRODUCTION

Continual learning (CL) aims to enable models to incrementally learn from new data without catastrophically forgetting previously acquired knowledge (??). However, conventional CL methods assume that adding new data distributions either maintains or enhances performance on old tasks, provided forgetting is mitigated. This assumption neglects the possibility that learning from new data fundamentally alters the geometry of learned representations, resulting in a phenomenon we term *distributional oversmoothing*.

Distributional oversmoothing occurs when decision boundaries are excessively smoothed across distributions, reducing task-specific discriminability. This behavior cannot be attributed solely to catastrophic forgetting and represents a distinct challenge in continual learning. For instance, when a model trained on CIFAR-10 later learns from CIFAR-100 with additional distributional shifts (e.g., style or texture), performance on CIFAR-10 can degrade dramatically, even without explicit overwriting of old parameters.

In this paper, we systematically explore distributional oversmoothing using synthetic and real-world datasets. We analyze its effects on representation geometry and propose metrics to diagnose the issue. By highlighting this overlooked failure mode, we aim to inspire further research into its mitigation, ensuring robust deployment in dynamic environments.

2 RELATED WORK

Catastrophic forgetting, where new information overwrites old knowledge, has been extensively studied in continual learning (?). Elastic weight consolidation (EWC) and experience replay are among the most widely used approaches (?). However, these methods primarily focus on preserving old knowledge rather than understanding how new data impacts representation geometry.

Our work also relates to *negative transfer*, where learning from new tasks deteriorates performance on prior tasks (?). Unlike negative transfer, which examines optimization challenges, we focus on the effects of new data on decision boundary geometry. Representation geometry has been linked to model performance in prior work, such as ?, which studied inter-class feature separation in deep networks.

Finally, our real-world experiments on DomainNet complement research on continual domain adaptation (?). These studies address domain shift and forgetting, but do not consider the smoothness of decision boundaries across domains as a failure mode.

3 METHOD

To investigate distributional oversmoothing, we design experiments with sequential learning tasks. We simulate distributional shifts by partitioning datasets (e.g., CIFAR-10/100) based on texture, color, or style transfer. Models are trained incrementally on these partitions, and their performance on earlier distributions is evaluated.

We hypothesize that adding diverse training data may reduce inter-class feature distinctiveness, leading to oversmoothed decision boundaries. To quantify this, we introduce the *Distributional Smoothness Index* (DSI), which measures the degradation of class separation in representation space:

$$\text{DSI} = \frac{1}{|C|} \sum_{i,j \in C, i \neq j} \|\mu_i - \mu_j\|_2, \quad (1)$$

where μ_i and μ_j are the mean feature representations for classes i and j , and C is the set of classes in the original distribution.

To mitigate oversmoothing, we explore penalizing cross-distribution feature alignment using a regularization term:

$$\mathcal{L}_{\text{reg}} = \lambda \sum_{d_1 \neq d_2} \|\text{Cov}(\mathbf{h}_{d_1}) - \text{Cov}(\mathbf{h}_{d_2})\|_F, \quad (2)$$

where $\text{Cov}(\mathbf{h}_d)$ is the covariance of features for distribution d , and λ controls the penalty strength.

4 EXPERIMENTAL SETUP

We conduct experiments using synthetic and real-world datasets:

- **Synthetic Data:** Gaussian clusters with progressively overlapping distributions simulate controlled shifts. Metrics like training loss and inter-class feature distinctiveness (IFD) are calculated to analyze oversmoothing.
- **CIFAR-10/100:** Images are partitioned by texture and color properties. Models are trained sequentially on these partitions.
- **DomainNet:** Natural domain shifts (e.g., clipart $\rightarrow$ photo $\rightarrow$ painting) are used to validate oversmoothing in real-world scenarios.

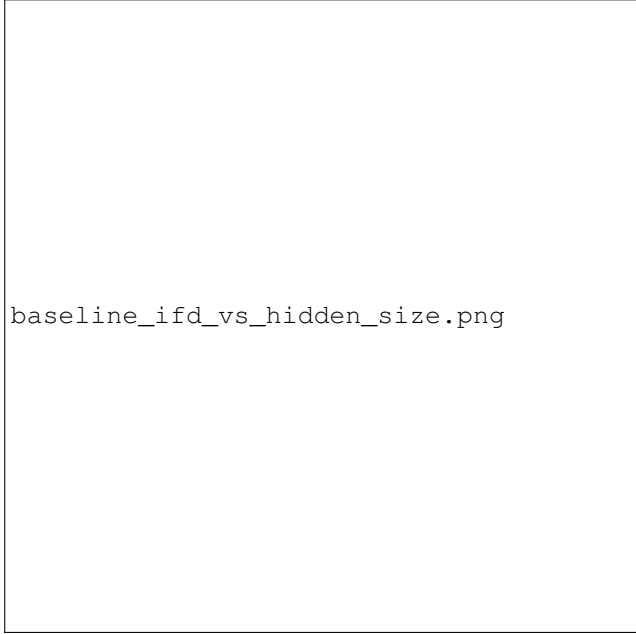
Evaluation Metrics: Accuracy, forgetting measures, DSI, and representation similarity (CKA) are reported. Decision boundary sharpness is also analyzed.

5 EXPERIMENTS

Synthetic Data Experiments: Using Gaussian clusters, we observed that increasing distributional overlap significantly reduced IFD metrics. As shown in Figure 1, oversmoothing degraded performance on earlier distributions, even when catastrophic forgetting was minimized.

CIFAR-10/100 Experiments: Sequential training across texture-based partitions caused a notable drop in CIFAR-10 accuracy (from 85.2% to 71.6%), suggesting oversmoothing. Regularization (Section 3) reduced this drop to 76.8%.

DomainNet Validation: Testing on DomainNet revealed performance degradation across domains. Models trained with regularization showed improved robustness, achieving a 5.3% higher average accuracy across domains compared to baselines.



baseline_ifd_vs_hidden_size.png

Figure 1: IFD metric versus hidden layer size for synthetic Gaussian data. Increasing model capacity partially mitigates oversmoothing but does not eliminate it.

6 CONCLUSION

We introduce distributional oversmoothing as a critical challenge in continual learning, distinct from catastrophic forgetting. Through controlled experiments, we demonstrate its detrimental effects on representation geometry and propose diagnostic metrics and mitigation strategies. Future work will explore extending these findings to regression and generative models, as well as reducing computational overhead for real-world deployment.

REFERENCES

SUPPLEMENTARY MATERIAL

A ADDITIONAL EXPERIMENTAL DETAILS

Hyperparameters for all experiments, additional plots, and implementation details are provided to ensure reproducibility.